INNOMATICS RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Music Store Data Analysis Using MYSQL

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About me

- **Background?**
- I completed my B.Tech in Data Science from Sreyas Institute of Engineering and Technology.

- **Why you want to learn Data Science**
- I want to learn Data Science because it helps solve real-world problems by analyzing data and making informed decisions.

- **Any work experience**
- I completed Data Analyst Intern Internship at Greenko Energies Private Limited where I gained hand-on experience in Python, SQL, Data analysis and preprocessing and working with real-time datasets.

- **Share your LinkedIn and GitHub profile URLs**
- www.linkedin.com/in/srivallijanani
- https://github.com/Srivalli-04

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Agenda

- **Business problem**
- **Business statement**
- **Introduction**
- **Data Description**
  - ER Diagram
  - Database Schema Overview
  - Cardinalities
- **Key analysis questions and insights**
- **Recommendations**
- **Conclusion**
- **Challenges**
- **Q&A Slide**

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Business Problem

- Difficulty in understanding customer purchasing and spending patterns.
- Unable to identify countries and cities generating higher sales.
- Unable to understand popular music genres across different countries.
- Difficulty in identifying top-performing artists and Rock music artists.
- Unable to identify high-value customers based on their spending.
- Unable to analyze track length and overall music purchasing behavior.

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BUSINESS STATEMENT

- Analyze sales and customer data to uncover revenue patterns, customer value, geographic performance, and music preferences.
- Design and implement a structured relational database with appropriate primary keys, foreign keys, and relationships.
- Import and validate Music Store data across the related tables to ensure data consistency and accuracy.
- Analyze customer purchases, invoices, artists, tracks, and genres using SQL queries.
- Apply SQL techniques such as joins, aggregations, subqueries, CTEs, filtering, and ranking to derive meaningful business insights.

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❑ INTRODUCTION

- Music Store Data Analysis is a SQL-based project using MySQL.
- The project works with data related to customers, employees, invoices, tracks, albums, artists, genres, and playlists.
- The data is organized into related relational tables using primary keys and foreign keys.
- The project uses SQL to analyze customer purchases, sales, artists, genres, and geographic patterns.
- The analysis helps convert the available Music Store data into meaningful business insights.

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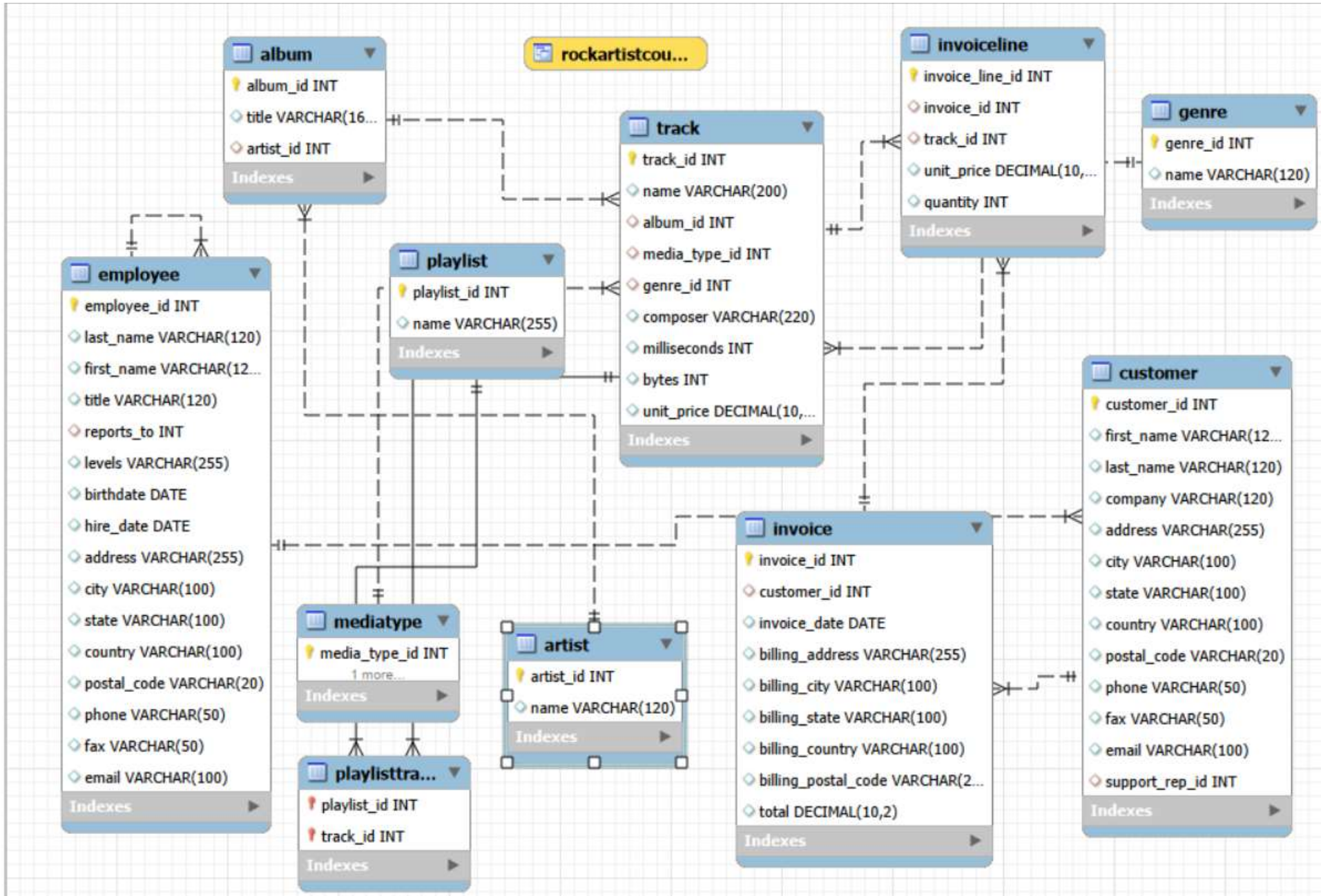
BUSINESS REQUIREMENTS

- Create and manage a structured Music Store database
- Import and validate the given CSV data
- Analyze customer and invoice data
- Analyze artists, tracks, and music genres
- Identify top customers, artists, and locations
- Use SQL to generate meaningful business insights

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DATA DESCRIPTION

- ER DIAGRAM



DATABASE SCHEMA OVERVIEW

Table Name	Description	Columns it Contains
Employee	Stores employee details, job roles, levels, and reporting relationships.	employee_id, last_name, first_name, title, reports_to, birthdate, hire_date, address, city, state, country, postal_code, phone, fax, email
Customer	Stores customer information and assigned support representative.	customer_id, first_name, last_name, company, address, city, state, country, postal_code, phone, fax, email, support_rep_id
Invoice	Records customer transactions, billing details, dates, and purchase totals.	invoice_id, customer_id, invoice_date, billing_address, billing_city, billing_state, billing_country, billing_postal_code, total
InvoiceLine	Stores individual tracks purchased within each invoice.	invoice_line_id, invoice_id, track_id, unit_price, quantity
Track	Stores individual music tracks and their associated metadata.	track_id, name, album_id, media_type_id, genre_id, composer, milliseconds, bytes, unit_price
Artist	Stores information about music artists.	artist_id, name
Album	Stores album information and its associated artist.	album_id, title, artist_id
Genre	Stores available music genre categories.	genre_id, name
MediaType	Stores the media types associated with tracks.	media_type_id, name
Playlist	Stores playlist information.	playlist_id, name
PlaylistTrack	Connects playlists with their associated tracks.	playlist_id, track_id

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CARDINALITIES

ParentTable	Child Table	Relationship	Cardinality
Artist	Album	One Artist can have many Albums	1 : N
Album	Track	One Album can have many Tracks	1 : N
Genre	Track	One Genre can have many Tracks	1 : N
MediaType	Track	One Media Type can have many Tracks	1 : N
Customer	Invoice	One Customer can have many Invoices	1 : N
Employee	Customer	One Employee can support many Customers	1 : N
Employee	Employee	One Employee can manage/report to many Employees	1 : N
Invoice	Invoice Line	One Invoice can contain many Invoice Lines	1 : N
Track	InvoiceLine	One Track can appear in many Invoice Lines	1 : N
Playlist	PlaylistTrack	One Playlist can have many PlaylistTrack records	1 : N
Track	PlaylistTrack	One Track can appear in many PlaylistTrack records	1 : N

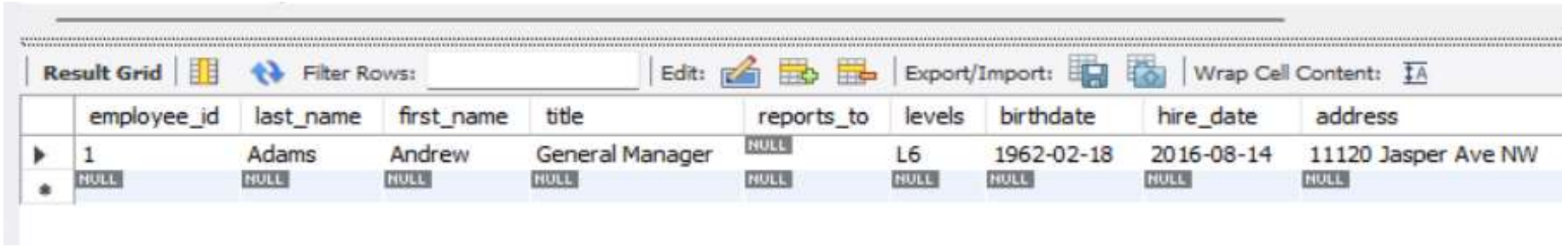
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❑ KEY ANALYSIS QUESTIONS

- Query 1 : **Who is the senior most employee based on job title?**

```
SELECT * FROM Employee
ORDER BY levels DESC
LIMIT 1;
```

- Output:



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	last_name	first_name	title	reports_to	levels	birthdate	hire_date	address
1	Adams	Andrew	General Manager	NULL	L6	1962-02-18	2016-08-14	11120 Jasper Ave NW
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Insight : Andrew Adams is the senior-most employee, with the highest level (L6) among the employees.

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- Query 2: **Which countries have the most Invoices?**

```sql
SELECT billing_country, COUNT(*) AS total_invoices
FROM Invoice
GROUP BY billing_country
HAVING COUNT(*) = (
    SELECT MAX(invoice_count)
    FROM (
        SELECT COUNT(*) AS invoice_count
        FROM Invoice
        GROUP BY billing_country
    ) AS country_counts
);
```

- Result:

Result Grid | Filter Rows:

	billing_country	total_invoices
▶	USA	131

- Insight: The analysis identifies the countries with the highest number of invoices, showing where the Music Store has the most customer transactions.

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- Query 3: **What are the top 3 values of total invoice?**

```
SELECT DISTINCT i1.total
FROM Invoice i1
LEFT JOIN Invoice i2
    ON i2.total > i1.total
GROUP BY i1.total
HAVING COUNT(DISTINCT i2.total) < 3;
```

- Result:

Result Grid

	total
▶	18.81
	19.80
	23.76

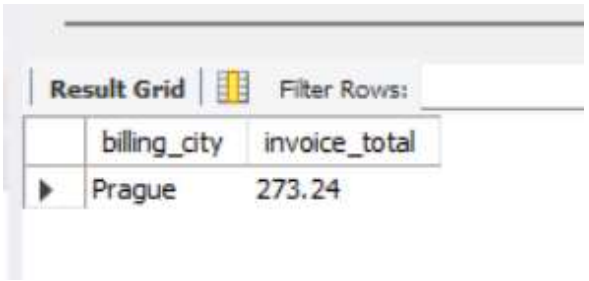
- Insight: This analysis identifies the three highest invoice amounts in the dataset. It helps highlight the store's largest individual transactions.

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- Query 4: **Which city has the best customers? - We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals**

```
WITH CitySales AS (
    SELECT billing_city, SUM(total) AS invoice_total
    FROM Invoice
    GROUP BY billing_city
),
RankedCities AS (
    SELECT billing_city, invoice_total,
           RANK() OVER (ORDER BY invoice_total DESC) AS city_rank
    FROM CitySales
)SELECT billing_city, invoice_total
FROM RankedCities
WHERE city_rank = 1;
```

- Result:



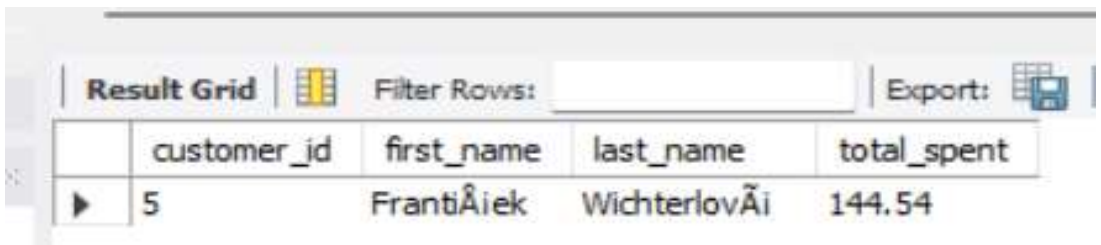
- Insight: The analysis identifies the city generating the highest total invoice revenue, helping the business identify its highest-revenue location.

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- Query 5: **Who is the best customer? - The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money**

```
WITH CustomerSpending AS (
    SELECT c.customer_id, c.first_name, c.last_name,
           SUM(i.total) AS total_spent FROM Customer c
    JOIN Invoice i ON c.customer_id = i.customer_id
    GROUP BY c.customer_id, c.first_name, c.last_name
),
RankedCustomers AS (
    SELECT customer_id, first_name, last_name, total_spent,
           DENSE_RANK() OVER (ORDER BY total_spent DESC) AS spending_rank FROM CustomerSpending
)
SELECT customer_id, first_name, last_name, total_spent FROM RankedCustomers WHERE spending_rank = 1;
```

- Result:



customer_id	first_name	last_name	total_spent
5	FrantiÅiek	WichterlovÃi	144.54

- Insight: The analysis identifies the customer who has spent the most money, representing the highest-value customer in the Music Store.

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- Query 6: **Write a query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A**

```
SELECT c.email, c.first_name, c.last_name, 'Rock' AS genre
FROM Customer c
WHERE EXISTS (
    SELECT 1
    FROM Invoice i
    JOIN InvoiceLine il ON i.invoice_id = il.invoice_id
    JOIN Track t ON il.track_id = t.track_id
    JOIN Genre g ON t.genre_id = g.genre_id
    WHERE i.customer_id = c.customer_id
      AND g.name = 'Rock'
)
ORDER BY c.email ASC;
```

- Result:



email	first_name	last_name	genre
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	Alexandre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	BjÃ¸rn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock

- Insight: The analysis identifies the customers who have purchased Rock music, helping the business understand its Rock music customer segment.

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- Query 7: **Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands**

```
CREATE VIEW RockArtistCounts AS
SELECT a.artist_id,
       a.name AS artist_name,
       COUNT(t.track_id) AS track_count
FROM Artist a
JOIN Album al ON a.artist_id = al.artist_id
JOIN Track t ON al.album_id = t.album_id
JOIN Genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY a.artist_id, a.name;
```

```
SELECT artist_name, track_count
FROM RockArtistCounts
ORDER BY track_count DESC
LIMIT 10;
```

- Output:

Result Grid | Filter Rows: | Expo

artist_name	track_count
AC/DC	18
Aerosmith	15
Audioslave	14
Led Zeppelin	14
Alanis Morissette	13
Alice In Chains	12
Frank Zappa & Captain Beefheart	9
Accept	4

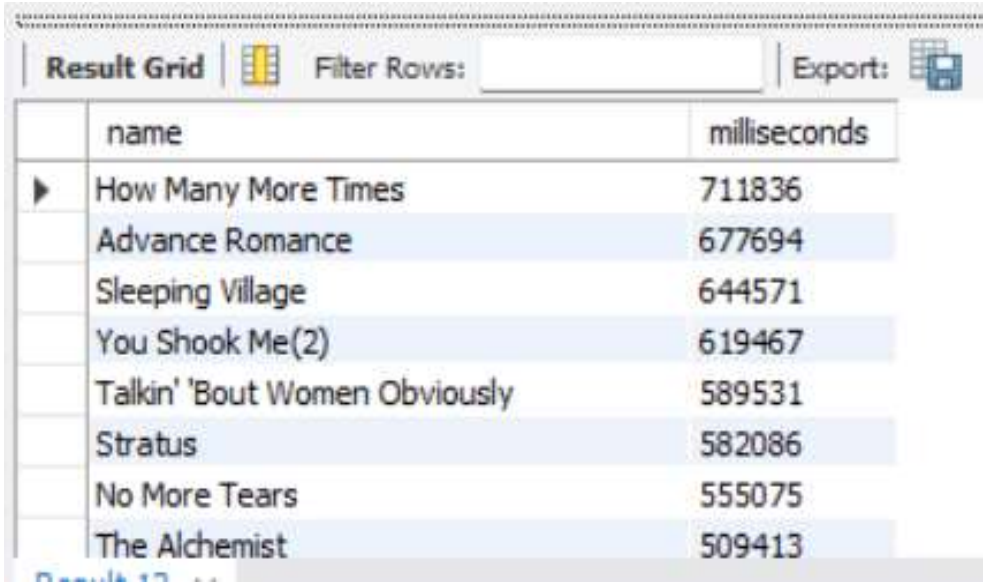
- Insight : The analysis identifies the top 10 artists with the highest number of Rock tracks in the Music Store dataset.

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- Query 8: **Return all the track names that have a song length longer than the average song length.- Return the Name and Milliseconds for each track. Order by the song length, with the longest songs listed first**

```
SELECT t.name, t.milliseconds
FROM Track t
CROSS JOIN (
    SELECT AVG(milliseconds) AS avg_milliseconds
    FROM Track
) AS avg_length
WHERE t.milliseconds > avg_length.avg_milliseconds
ORDER BY t.milliseconds DESC;
```

Output:



Result Grid | Filter Rows: | Export:

name	milliseconds
How Many More Times	711836
Advance Romance	677694
Sleeping Village	644571
You Shook Me(2)	619467
Talkin' 'Bout Women Obviously	589531
Stratus	582086
No More Tears	555075
The Alchemist	509413

- Insight : The analysis identifies tracks that are longer than the average song length and highlights the longest tracks in the Music Store dataset.

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- Query 9: **Find how much amount is spent by each customer on artists? Write a query to return customer name, artist name and total spent**

```sql
SELECT
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    a.name AS artist_name,
    SUM(il.unit_price * il.quantity) AS total_spent
FROM Customer c
JOIN Invoice i
    ON c.customer_id = i.customer_id
JOIN InvoiceLine il
    ON i.invoice_id = il.invoice_id
JOIN Track t
    ON il.track_id = t.track_id
JOIN Album al
    ON t.album_id = al.album_id
JOIN Artist a
    ON al.artist_id = a.artist_id
GROUP BY
    c.customer_id,
    c.first_name,
    c.last_name,
    a.artist_id,
    a.name
ORDER BY total_spent DESC;
```

Output:

Result Grid | Filter Rows: | Export:

customer_name	artist_name	total_spent
Steve Murray	AC/DC	17.82
Jennifer Peterson	Aerosmith	14.85
Mark Taylor	Aerosmith	14.85
Leonie KÃ¶hler	Audioslave	13.86
Fernanda Ramos	AntÃ´nio Carlos Jobim	13.86
Edward Francis	Alanis Morissette	12.87
JoÃ£o Fernandes	Alanis Morissette	12.87
Emma Jones	Alanis Morissette	12.87
Victor Stevens	Alice In Chains	11.88
Kathy Chase	AC/DC	10.89

- Insight: This analysis shows how much each customer spends on different artists. It helps understand individual customer preferences and spending patterns across artists.

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- Query 10: **We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared, return all Genres**

```
DELIMITER //
CREATE PROCEDURE MostPopularGenreByCountry()
BEGIN SELECT country,genre,total_purchases FROM (
        SELECT i.billing_country AS country,g.name AS genre,
            SUM(il.quantity) AS total_purchases FROM Invoice i
        JOIN InvoiceLine il
            ON i.invoice_id = il.invoice_id
        JOIN Track t
            ON il.track_id = t.track_id
        JOIN Genre g
            ON t.genre_id = g.genre_id
        GROUP BY i.billing_country, g.genre_id, g.name
    ) AS genre_totals WHERE (country, total_purchases) IN ( SELECT country, MAX(total_purchases) FROM (
            SELECT i.billing_country AS country,g.name AS genre,SUM(il.quantity) AS total_purchases
            FROM Invoice i JOIN InvoiceLine il
                ON i.invoice_id = il.invoice_id
            JOIN Track t
                ON il.track_id = t.track_id
            JOIN Genre g
                ON t.genre_id = g.genre_id GROUP BY i.billing_country, g.genre_id, g.name) AS country_totals
        GROUP BY country
    ) ORDER BY country;
END //
DELIMITER ;
CALL MostPopularGenreByCountry();
```

Result :

country	genre	total_purchases
Argentina	Rock	1
Australia	Rock	18
Austria	Rock	6
Belgium	Rock	5
Brazil	Rock	26
Canada	Rock	57
Chile	Rock	7
Czech Republic	Rock	14
Denmark	Rock	6
Finland	Rock	6
France	Rock	26
Germany	Rock	28
Hungary	Rock	4
India	Rock	13
Ireland	Rock	2
Italy	Rock	3

- Insight: This query identifies the most purchased genre in each country. It helps understand differences in music preferences across countries and also identifies cases where multiple genres have equal popularity.

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❑ RECOMMENDATIONS:

- **Interactive Visualizations** – Build dynamic dashboards with charts, filters, and geographic views to explore sales, customer, and music trends.
- **Customer Segmentation** – Group customers based on their purchasing behavior and spending patterns to support targeted marketing strategies.
- **Sales Forecasting** – Predict future revenue and music demand using historical sales and invoice data.
- **Churn Prediction** – Identify customers whose purchasing activity is declining and develop strategies to improve customer retention.
- **Real-Time Analytics** – Connect the Music Store database to a live dashboard to monitor continuously updated sales and customer KPIs.

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CONCLUSION:

The Music Store analysis provided a clear understanding of how customer transactions and music-related data can be explored through SQL. The project helped uncover key patterns in customer spending, sales across locations, popular genres, artist performance, and purchasing behavior. Overall, the analysis shows how organizing data in a relational database and applying SQL techniques can turn large volumes of store data into useful insights for better business planning and customer-focused decisions.

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Question and Answer



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THANK YOU



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